

Cryptographically Auditable Decision Governance for Autonomous Trading Systems: An Early Production Deployment Study

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Abstract

We present an early production deployment of a fail-closed decision governance framework for autonomous trading systems, combining a six-checkpoint evaluation pipeline with post-quantum cryptographic audit trails. Over 49 days of continuous operation, the system processed 693,890 evaluation cycles and generated 27,449 governance receipts, each cryptographically signed using Dilithium-3 (NIST-standardized ML-DSA-65) and linked into a SHA-256 hash chain providing tamper-evident provenance. Unlike prior governance frameworks that operate as post-hoc compliance layers, our architecture enforces governance at decision time: any gate failure produces an immediate BLOCKED outcome with no override path. The system is domain-agnostic — the same six-checkpoint pipeline has been validated for trading signals and extended via a REST API to credit and insurance domains. All receipts are publicly verifiable without access to internal systems. To our knowledge, this is the first reported early production deployment of post-quantum cryptographic governance receipts for autonomous decision systems.

Keywords: decision governance, autonomous systems, post-quantum cryptography, audit trail, fail-closed architecture, algorithmic trading

1. Introduction

On August 1, 2012, Knight Capital Group deployed a trading algorithm that executed 4 million transactions in 45 minutes before human operators could intervene, generating a loss of approximately \$440 million — effectively bankrupting the firm. The incident has since served as a canonical example of what happens when autonomous systems operate without real-time governance constraints.

Existing responses to this problem fall into two categories: pre-trade risk limits (circuit breakers, position limits) and post-trade compliance reporting. Both are necessary but insufficient. Pre-trade limits are often static and fail to account for dynamic market conditions. Post-trade reporting is retrospective — it documents what went wrong after the loss has occurred.

We propose a third approach: **decision governance as infrastructure**, applied at evaluation time, before any action is executed, with cryptographic proof of every decision embedded in an append-only chain. This paper reports on an early production deployment of this approach in an autonomous cryptocurrency trading system, operated continuously from January 9 to February 27, 2026.

The system processed **693,890 evaluation cycles** across this period, generating **27,449 governance receipts**, each carrying a Dilithium-3 post-quantum digital signature and a SHA-256 link to all prior receipts. Every receipt is publicly verifiable at an open endpoint — an auditor or regulator with no internal system access can independently confirm the authenticity and integrity of any governance decision.

2. Background and Related Work

2.1 Algorithmic Trading Governance

Regulatory frameworks such as MiFID II in the EU and FINRA Rule 3110 in the US require firms to maintain audit trails for algorithmic trading decisions. However, these regulations specify what must be recorded, not how to enforce governance at decision time. The result is systems that are compliant in documentation but ungoverned in execution.

Recent work in AI governance proposes frameworks for transparency and accountability in automated decision systems, primarily through explainability requirements and human oversight provisions. These frameworks are architecturally passive — they do not prevent a decision from executing; they document it afterward.

2.2 Cryptographic Audit Trails

Hash-chained audit logs have been studied extensively in database systems and are the basis of blockchain architectures. Applied to governance receipts, a hash chain provides the property that any modification to a historical record is mathematically detectable. This transforms the audit trail from a compliance artifact into a cryptographic commitment.

2.3 Post-Quantum Cryptography

NIST finalized three post-quantum cryptographic standards in 2024: ML-KEM (FIPS 203), ML-DSA (FIPS 204), and SLH-DSA (FIPS 205). Of these, ML-DSA (also known as Dilithium) provides digital signature functionality. To our knowledge, the application of these standards to real-time governance receipt signing in production autonomous systems has not previously been reported.

3. System Architecture

3.1 Overview

The governance framework consists of three components: (1) a six-checkpoint evaluation engine, (2) a receipt generation and signing module, and (3) a public verification endpoint. These components operate as a pipeline — every decision request enters through the evaluation engine and exits through the receipt module, regardless of the evaluation outcome.

3.2 Six-Checkpoint Evaluation Pipeline

All signals are normalized to a [0, 100] scale. The six checkpoints evaluate orthogonal dimensions of decision quality:

Checkpoint	Signal	Pass Condition	Rationale
CP-1 Probability	probability_score	≥ 50	Minimum positive expected outcome
CP-2 Risk Limits	risk_exposure	≤ 65	Maximum tolerable exposure
CP-3 Signal Coherence	signal_coherence	≥ 55	Internal signal agreement
CP-4 Trend Persistence	trend_persistence	≥ 50	Temporal regime stability
CP-5 Stress Resilience	stress_resilience	≥ 35	Adverse scenario survival
CP-6 Logic Consistency	logic_consistency	≥ 40	Absence of internal contradiction

Fail-closed property: a single gate failure produces a BLOCKED decision with no escalation path. This is architecturally distinct from scoring systems where a low score on one dimension can be compensated by a high score on another.

3.3 Hierarchical Veto Flow

Prior to the six-checkpoint gate, the system applies four sequential veto mechanisms operating on market-specific signals:

1. **Monte Carlo VETO** — simulates trade outcomes; blocks if win rate < 50% or expected return $\leq 0\%$
2. **RMS VETO** — risk management system constraints
3. **Adaptive Coherence Gate** — dynamic signal quality threshold
4. **Edge Confirmation Window (ECW)** — requires edge persistence across N consecutive cycles before permitting action

3.4 Post-Quantum Governance Receipts

Each evaluation generates a governance receipt containing a unique receipt ID, ISO 8601 UTC timestamp, asset identifier, decision (APPROVED / BLOCKED / HOLD), full veto chain log, policy version, SHA-256 hash of the prior receipt, SHA-256 hash of the current receipt payload, and a Dilithium-3 (ML-DSA-65) digital signature over the content hash. The prev_hash field links each receipt to its predecessor, forming an append-only chain verifiable by any external party.

4. Production Results

4.1 Operational Period

The system operated continuously from **January 9 to February 27, 2026** — 49 days, 3 hours of uninterrupted autonomous operation with a 90-second evaluation interval.

Metric	Value
Total evaluation cycles	693,890
Average cycles per day	~14,161
January 2026 cycles	279,252
February 2026 cycles	414,638
Growth Jan→Feb	+48.4%
PQC-signed governance receipts	27,449
Signature coverage	100%
Hash chain breaks	0

4.2 Market Regime During Period

The evaluation period coincided with a predominantly bearish market regime. EMA Regime Signal distribution across 693,890 cycles:

Signal	Count	Percentage
SHORT (bearish)	503,394	72.5%
NONE (neutral)	147,314	21.2%
LONG (bullish)	43,186	6.2%

4.3 Governance Decision Distribution

Decision	Count	Percentage
HOLD	27,439	99.96%
APPROVED	6	0.02%
BLOCKED	2	0.007%

The high HOLD rate is the expected behavior of the Edge Confirmation Window under the observed market regime. In a market where the primary signal is SHORT 72.5% of the time, the conditions for a LONG position to satisfy all six checkpoints simultaneously are rare by design. The system is not failing to trade — it is correctly withholding capital during an unfavorable regime.

4.4 Veto Chain Analysis — Representative Examples

Example 1 — HOLD (ECW pending):

MC_VETO: PASS (WR=52%), COHERENCE_GATE: PASS (67%), ECW: PENDING (cycle 1/3)

Example 2 — BLOCKED (insufficient win rate):

MC_VETO: BLOCKED (WR=43%, ER=-0.2%), RMS: N/A

Example 3 — APPROVED:

MC_VETO: PASS (WR=55%, ER=+0.8%), COHERENCE_GATE: PASS (72%), ECW: PASS (3/3 cycles)

5. Domain-Agnostic Extension

The six-checkpoint pipeline is architecturally domain-agnostic. The same governance mechanism has been formalized as a REST API endpoint accepting normalized signals (0–100) from any external system. A credit decisioning engine, an insurance underwriting platform, or an energy trading system submitting normalized signals receives the same cryptographic governance proof as the internal trading system.

Signal	Trading	Credit	Insurance
probability_score	MC win probability	Repayment probability	Claim frequency
risk_exposure	Portfolio VaR	Debt-to-income ratio	Exposure concentration
signal_coherence	EMA/HMM agreement	Score model consensus	Underwriting consistency
trend_persistence	Regime stability	Payment history	Loss history stability
stress_resilience	Black swan resilience	Adverse scenario survival	Catastrophe resilience
logic_consistency	DCI (inverted)	Scorecard consistency	Policy logic integrity

6. Data Governance

The production system implements client-level data isolation from inception. Each API request includes a client identifier stored alongside the governance receipt, enabling per-client audit queries without cross-client data exposure. Client signal payloads are encrypted at the application layer prior to storage using Fernet (AES-128-CBC with HMAC-SHA256 integrity verification). Governance receipts carry a retention timestamp set at ingestion time (creation date + 365 days, configurable per contract).

7. Discussion

7.1 Governance as Infrastructure vs. Compliance Layer

The conventional approach to algorithmic governance treats compliance as a layer applied over an existing system — an audit log, a reporting mechanism, a risk limit checker. Our architecture inverts this relationship: governance is the infrastructure through which every decision must pass. There is no path from signal to action that bypasses the evaluation pipeline. A fail-closed pipeline where any gate failure produces a BLOCKED outcome is architecturally incapable of being overridden.

7.2 Cryptographic Verifiability as a Design Principle

Standard audit trails are authoritative only insofar as the organization maintaining them is trusted to have recorded events accurately. The hash-chained, PQC-signed receipt architecture provides a materially different guarantee: any external party can verify that a specific governance decision occurred at a specific time with specific parameters, and that no receipt in the chain has been modified

or deleted since generation.

7.3 Post-Quantum Readiness

The transition from classical to post-quantum cryptographic standards is underway globally. NIST finalized the first PQC standards in 2024; regulatory adoption is expected to follow within 3–7 years. By deploying ML-DSA-65 in production in 2025, the system establishes a governance audit trail that will remain cryptographically valid as the threat landscape evolves. Governance receipts signed with classical ECDSA would require re-signing or could become inadmissible as quantum computing capabilities mature.

7.4 Limitations

Several limitations of the current early deployment should be noted: (1) The 49-day observation window is a short production window; longer-term data is needed to validate behavior across full market cycles. (2) Checkpoint thresholds are uniform across all clients — per-client configurable thresholds are planned. (3) Signal normalization is client-side: the system evaluates normalized signals but does not validate the normalization process itself. (4) Counterfactual analysis is partial: of 693,890 evaluation cycles, verified counterfactual outcomes are available for 50 events. Expanding this coverage is a priority for future work. (5) Single-node deployment: distributed consensus mechanisms are not yet implemented.

7.5 External Reproducibility Note

Any external party can independently verify the governance receipts reported in this study without access to internal systems, source code, or proprietary data. The following endpoints are publicly available on the production server with no authentication required:

Interactive verification:

```
https://omnibotgenesis-production.up.railway.app/verify
```

Enter any receipt_id in the search field to retrieve the full receipt, inspect the veto chain, validate the hash chain link to the preceding receipt, and confirm signature status.

Programmatic verification:

```
GET /api/verify/{receipt_id}
```

Returns a JSON object containing: receipt_id, decision, asset, timestamp, veto_chain, prev_hash, content_hash, signature, signature_algorithm, and a verification sub-object with boolean fields hash_valid and signature_valid.

Public key retrieval:

```
GET /api/public_key
```

Returns the Dilithium-3 (ML-DSA-65) public key in base64 encoding. Any conforming implementation of the ML-DSA-65 standard can use this key to independently verify the signature field of any receipt without trusting the server.

Fields to validate for full verification:

Field	Expected Value	What It Proves
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hash_valid	true	content_hash matches the stored receipt payload
signature_valid	true	Dilithium-3 signature over content_hash is authentic
prev_hash	matches prior receipt	Chain continuity — no receipt deleted or reordered
signature_algorithm	Dilithium-3 (ML-DSA-65)	NIST-standardized PQC signature scheme

Receipt ID format: OMNIX-{12-char hex} (e.g., OMNIX-a3f7c2b1d904). No API key is required for any verification endpoint. No internal system access is required.

8. Conclusion

We have described and reported on an early production deployment of a fail-closed, cryptographically auditable decision governance system for autonomous trading. Over 49 days of operation, the system generated 693,890 evaluation cycles and 27,449 governance receipts, each signed with Dilithium-3 post-quantum signatures and linked in an unbroken SHA-256 hash chain publicly verifiable without access to internal systems.

The architecture demonstrates that governance at decision time — rather than as a post-hoc compliance layer — is practically implementable with commercially available infrastructure. The domain-agnostic formulation of the six-checkpoint pipeline extends this guarantee to any automated decision system capable of expressing its decision signals on a normalized scale.

The primary contribution of this work is not a novel algorithm — it is a novel integration: fail-closed governance, cryptographic auditability, and post-quantum signature standards applied to the specific problem of autonomous system accountability. As autonomous decision systems proliferate across financial services, healthcare, energy, and transportation, the infrastructure to hold them accountable becomes as important as the systems themselves. This early deployment establishes a foundation for longer-term validation across full market cycles.

The governance receipts generated by this system from January 9, 2026 onward are publicly verifiable at:

<https://omnibotgenesis-production.up.railway.app/verify>

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